Data Analysis Report

Dataset: Sentiment / Social Media Engagement Data
Tool Used: Microsoft Excel
Objective: To analyze engagement patterns across platforms, time periods, and regions in order to generate actionable business insights for decision makers.

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1. Overview of the Analysis

This analysis evaluates social media engagement across 732 posts using metrics such as **likes, retweets, post volume, timing, geography, and trends over time**. The goal is to identify where engagement is strongest, when users are most active, and which strategies can maximize impact.

2. **Descriptive Statistics & Correlation Analysis**

DESCRIPTIVE STATISTICS		
Statistic	Retweets	Likes
Mean	21.51	42.90
Median	22	43
Mode	22	45
Minimum	5	10
Maximum	40	80
Variance	49.86	198.52
Standard Deviation	7.06	14.09
Count	732	732

CORRELATION & COVARIANCE	
Purpose	Value(Likes&Retweets)
Correlation	0.998
Covariance	99.34

Observation

Average likes (42.9) are higher than retweets (21.5), and there is a **very strong positive correlation (0.998)** between likes and retweets.

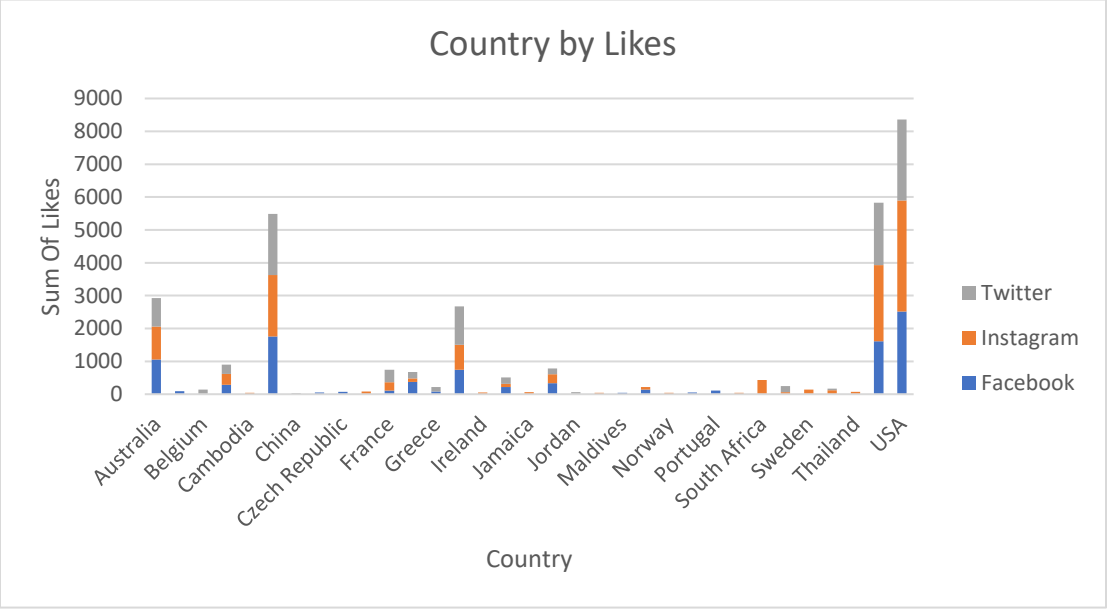
Interpretation

Content that attracts likes is very likely to also generate retweets, indicating consistent engagement behavior.

Recommendation

Focus on strategies that increase **likes**, as this will naturally boost overall engagement.

3.



Observation

The **USA** records the highest number of likes, followed by countries such as UK and Canada. Platform performance varies by country.

Interpretation

Geographic differences influence platform effectiveness and user behavior.

Recommendation

Prioritize high-performing countries and tailor platform strategies to regional preferences.

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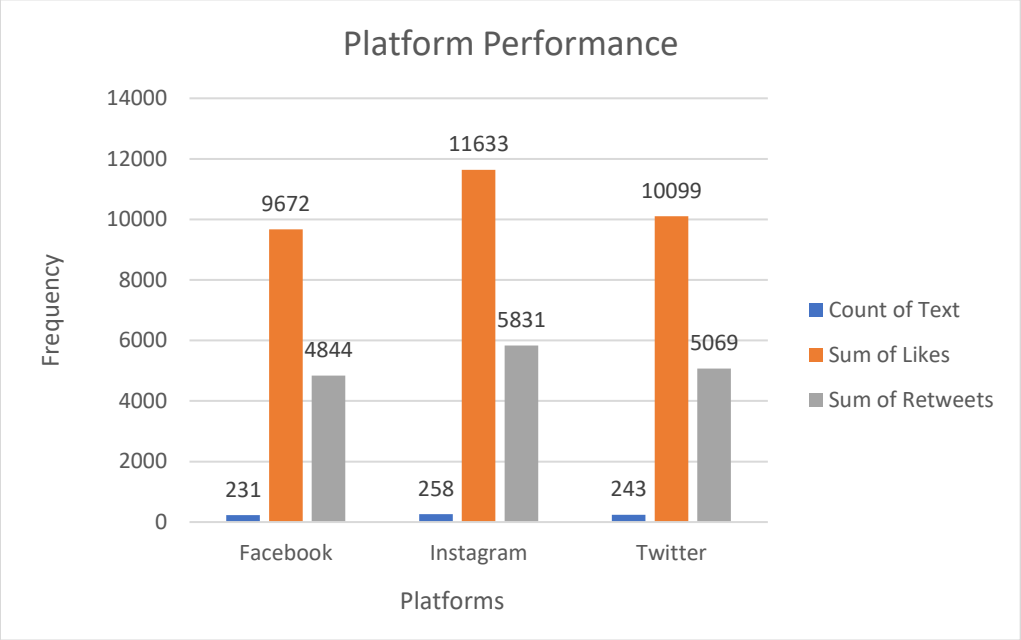

Observation

Instagram records the highest engagement overall, with the most likes and retweets, followed by Twitter. Facebook shows the lowest engagement and post volume.

Interpretation

Instagram is the most effective platform for audience interaction, while Facebook underperforms comparatively.

Recommendation

Management should prioritize **Instagram-led campaigns** and review Facebook's content strategy to improve engagement.

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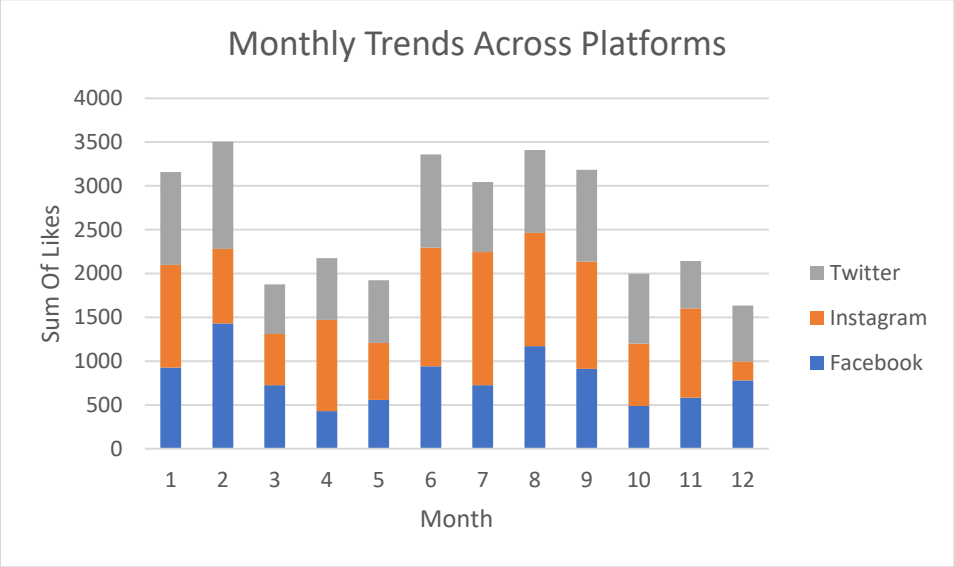

Observation

Engagement fluctuates throughout the year, with peaks around **months 1,2, 6, 9, and 8**. Instagram consistently contributes the largest share of monthly engagement.

Interpretation

Engagement is seasonal, and Instagram remains the primary driver of performance across months.

Recommendation

Schedule major campaigns during **high-engagement months** and analyze low-performing periods to improve content planning.

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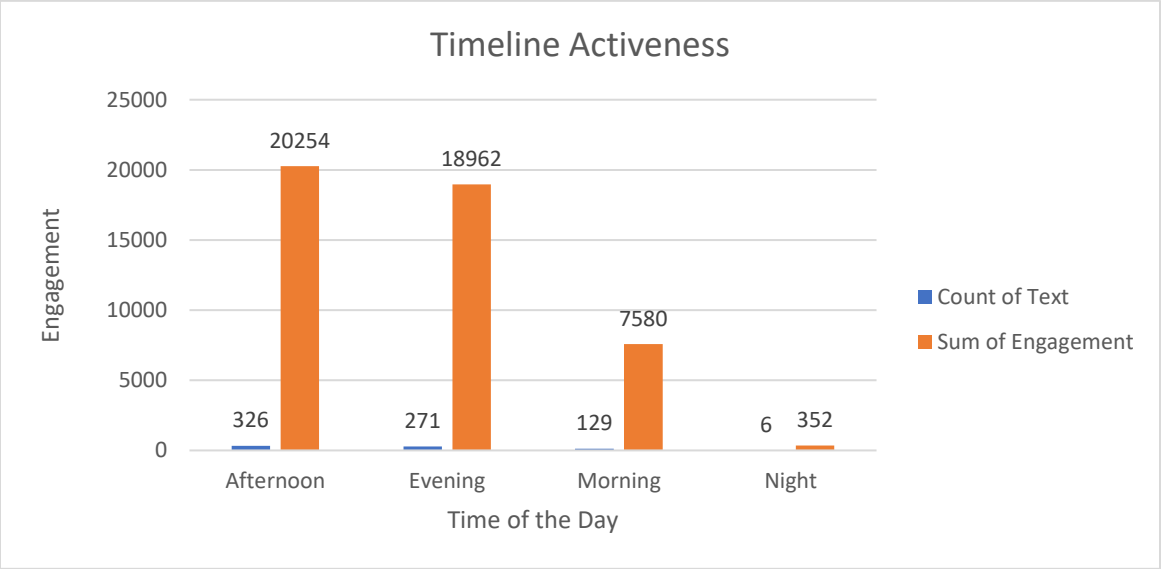

Observation

Engagement is highest in the **afternoon and evening**, moderate in the morning, and lowest at night.

Interpretation

User activity is concentrated later in the day, making timing a key factor in content visibility.

Recommendation

Post content mainly in the **afternoon and evening**, and limit posting during nighttime hours.

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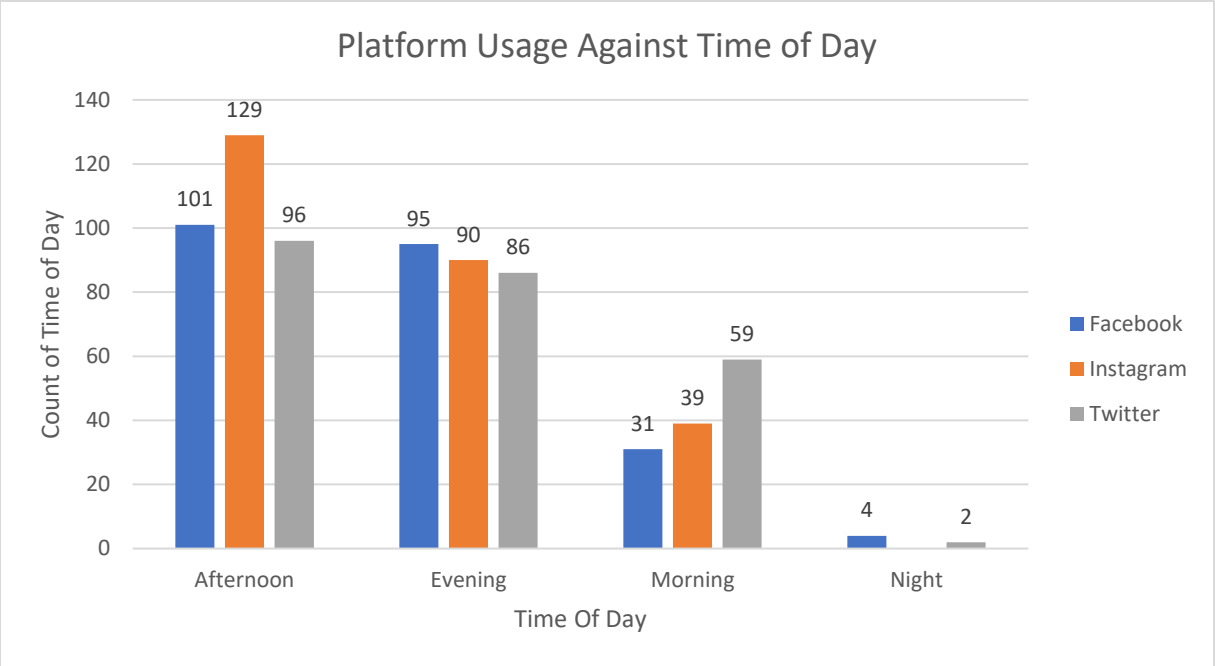

Observation

All platforms peak in the afternoon, with Instagram leading at every time period. Engagement drops significantly at night.

Interpretation

User behavior is consistent across platforms, and Instagram maintains dominance regardless of time.

Recommendation

Align posting schedules across platforms to **peak activity periods**, with a stronger focus on Instagram.

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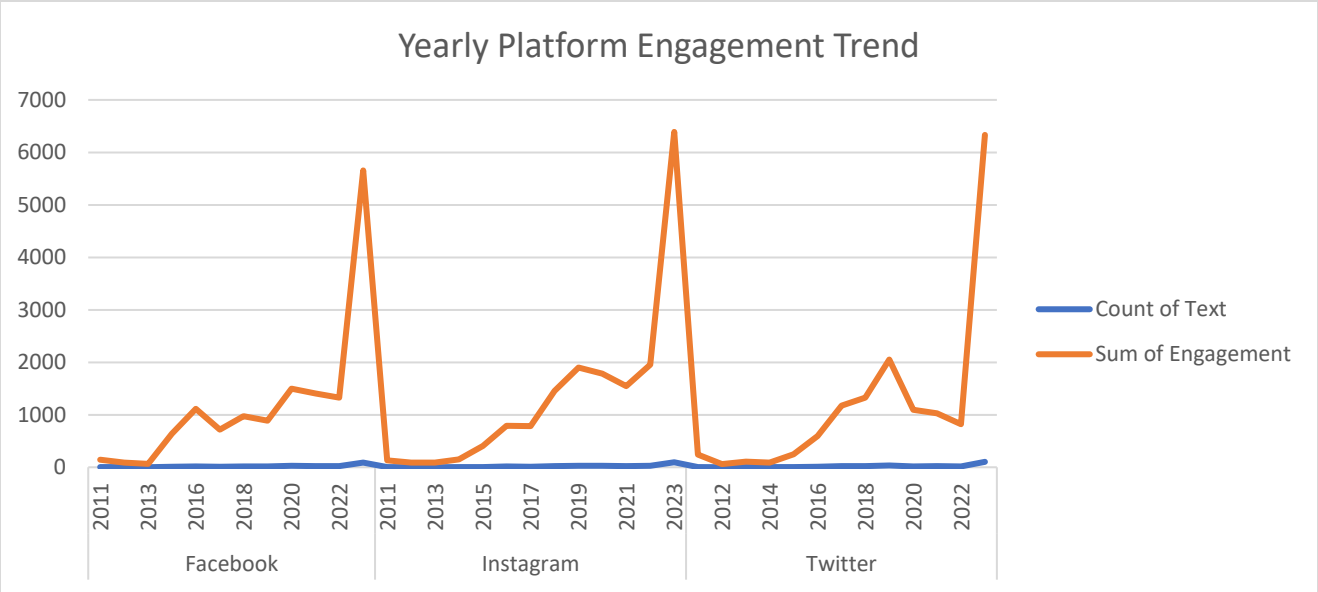

Observation

Engagement shows sharp spikes in certain years, while the number of posts remains relatively stable.

Interpretation

High engagement is driven more by **content effectiveness and campaign timing** than posting frequency.

Recommendation

Identify and replicate strategies used during **high-engagement years** to improve future performance.

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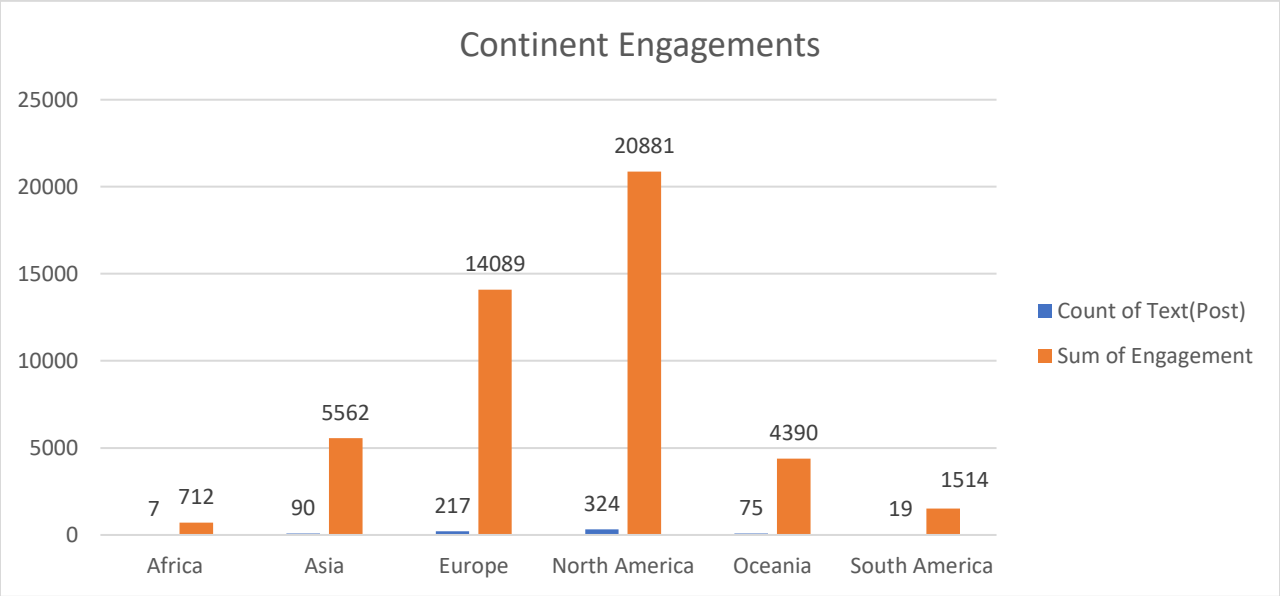

Observation

North America has the highest engagement, followed by **Europe**, while other continents show significantly lower interaction.

Interpretation

Audience response is concentrated in a few key regions.

Recommendation

Focus resources on **North America and Europe**, while developing targeted growth strategies for underperforming regions

KEY FINDINGS SUMMARY

- **Instagram is the top-performing platform**, generating the highest likes and retweets across all periods.
- **Engagement peaks in the afternoon and evening**, with significantly lower activity at night.
- **Mid-year months** record the highest engagement, indicating seasonal user behavior.

- **North America and Europe**, especially the **USA**, contribute the largest share of engagement.
- **Content impact and timing** drive engagement more than posting volume.
- **Likes and retweets are highly correlated (0.998)**, showing consistent audience response across engagement types.

BUSINESS RECOMMENDATIONS

1. Prioritize **Instagram-focused campaigns** to maximize engagement and reach.
2. Schedule posts primarily during **afternoon and evening hours** to align with peak user activity.
3. Plan major campaigns around **mid-year high-engagement periods**.
4. Concentrate marketing resources on **North America and Europe**, while testing localized strategies in underperforming regions.
5. Optimize content to drive **likes**, as this will also increase retweets and overall engagement.

CONCLUSION

Data-driven insights show that social media engagement is primarily influenced by **platform choice, timing, and content quality** rather than posting frequency. By focusing on high-performing platforms, peak activity periods, and key regions, the organization can significantly improve engagement efficiency and campaign effectiveness.